THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · EASY LEVEL CHAPTER TEST SET 2

Chapter 4: **Electricity: Magnetic and Heating Effects** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Oersted's discovery relates to: 1

(a) light and plants **(b)** air and water
(c) electricity and magnetism **(d)** heat and sound

2. To make an electromagnet we need an iron nail and: 1

(a) plastic **(b)** insulated wire and a cell
(c) water **(d)** paper

3. A switch is used to: 1

(a) make electricity **(b)** store charge
(c) measure current **(d)** start or stop current

4. Compass needle is a: 1

(a) wire **(b)** bulb **(c)** switch **(d)** small magnet

5. In an electric bell, the iron piece is attracted by: 1

(a) a compass **(b)** a cell **(c)** an electromagnet **(d)** a bulb

6. Electric current is measured in: 1

(a) kilograms **(b)** amperes **(c)** metres **(d)** seconds

7. The heater element is made of: 1

(a) gold **(b)** nichrome **(c)** iron nail **(d)** silver

8. A battery is made of: 1

(a) wires **(b)** bulbs **(c)** several cells **(d)** switches

9. An electromagnet works when: 1

(a) core is wooden **(b)** no current flows **(c)** current flows **(d)** wire is cut

10. A wire carrying current is surrounded by: 1

(a) sound **(b)** water **(c)** a magnetic field **(d)** oil

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. What does a switch do? 2

12. Which effect of current is used in an electric bell? 2

13. Name two insulators. 2

14. State the unit of electric current. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. List the symbols of cell, switch and bulb. 3

16. How can the strength of an electromagnet be increased? 3

17. Name three effects of electric current. 3

OR

Write three uses of electromagnets. 3

18. What does the compass needle do near a current wire? 3

OR

Write three points about an electric cell. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe how to make an electromagnet. 5

OR

State five uses of electricity in your home using different effects. 5

20. Explain Oersted's experiment. 5

OR

Compare an electromagnet and a bar magnet in five points. 5

END OF QUESTION PAPER Total: 40 Marks

MASTERMIND